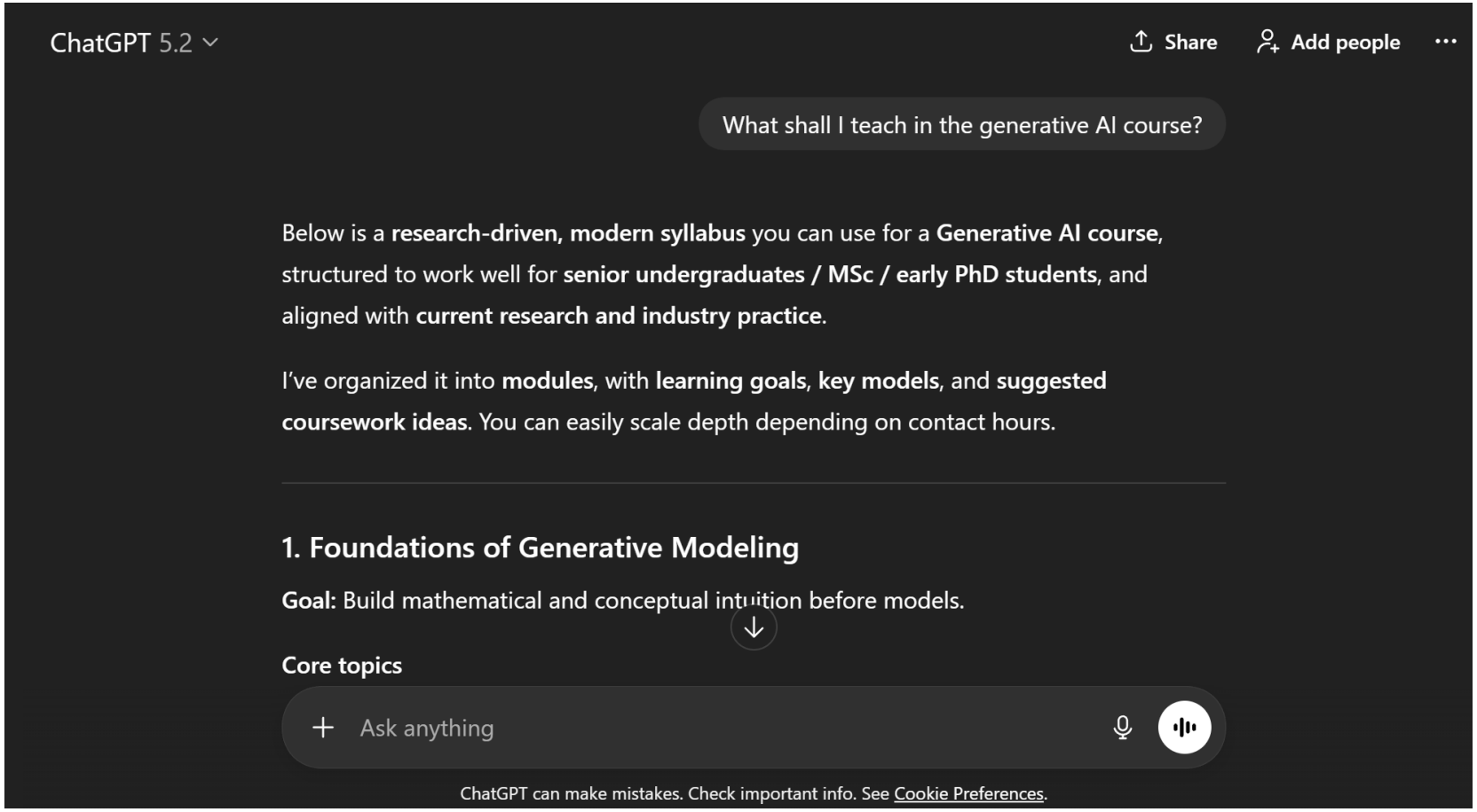
Introduction to Generative AI

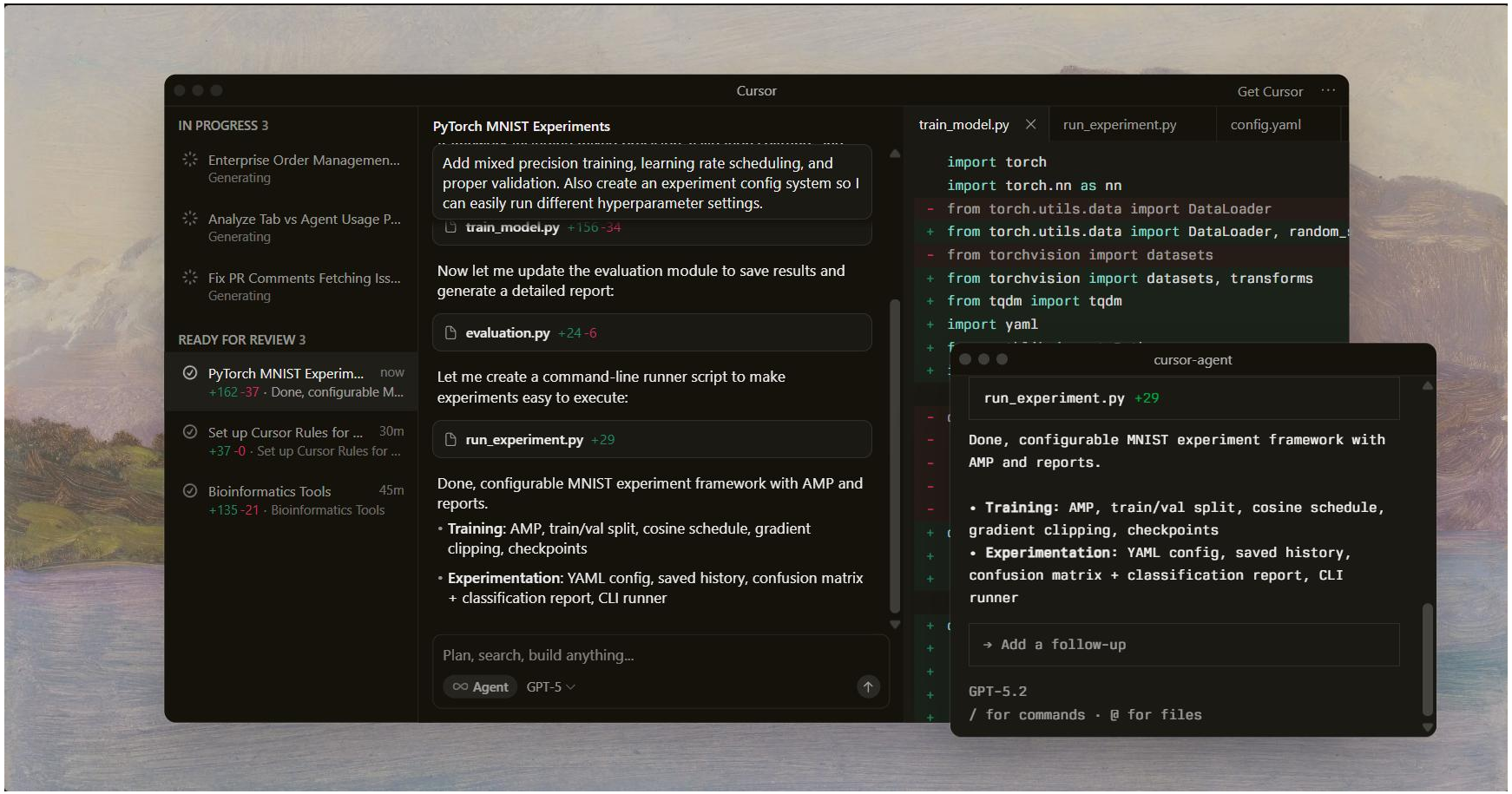
70113 Generative AI

Jiankang Deng
Spring 2026, Computing, Imperial College

Chatbot and Natural Language Conversation



Code Generation



Audio Generation

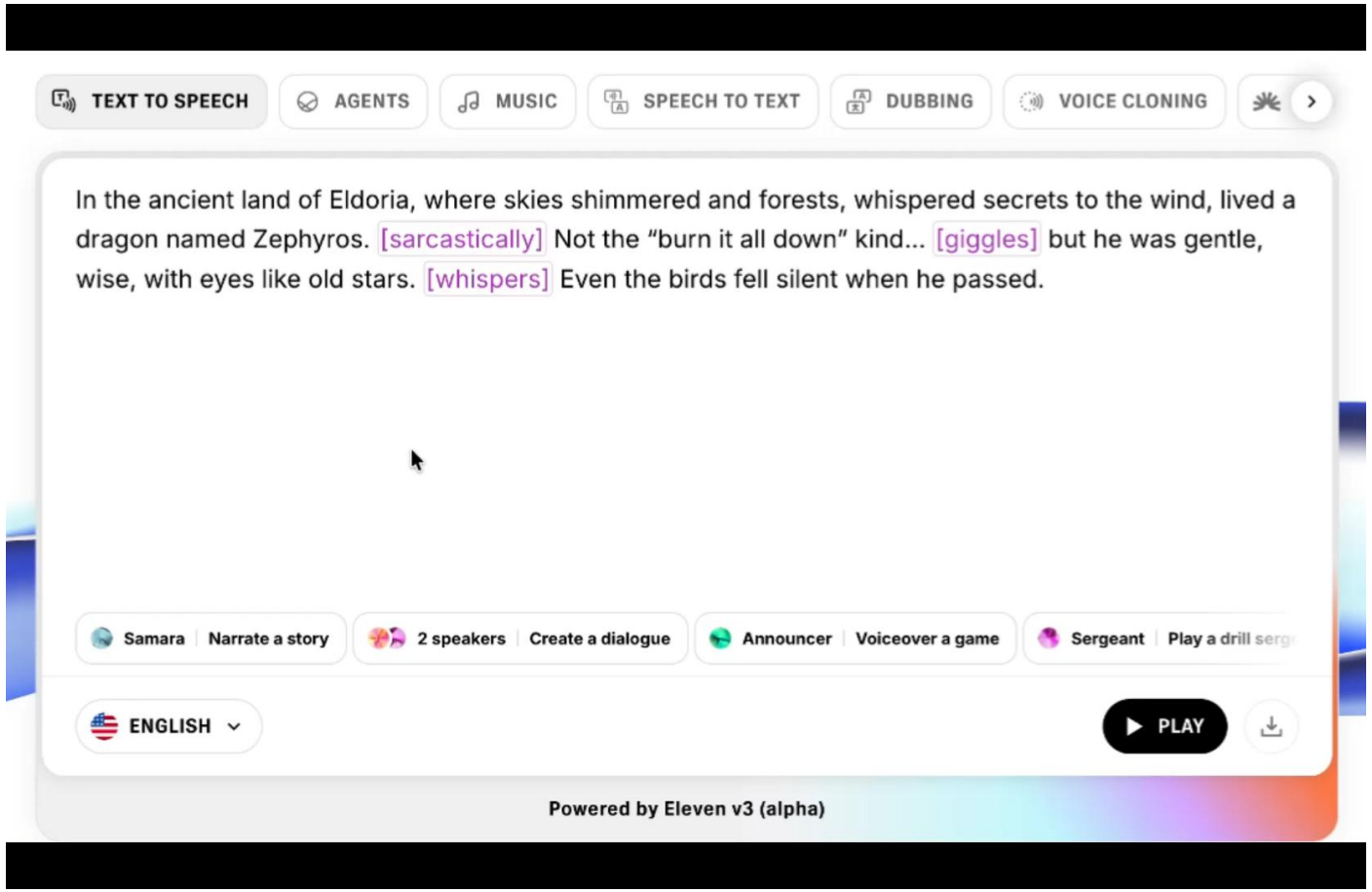
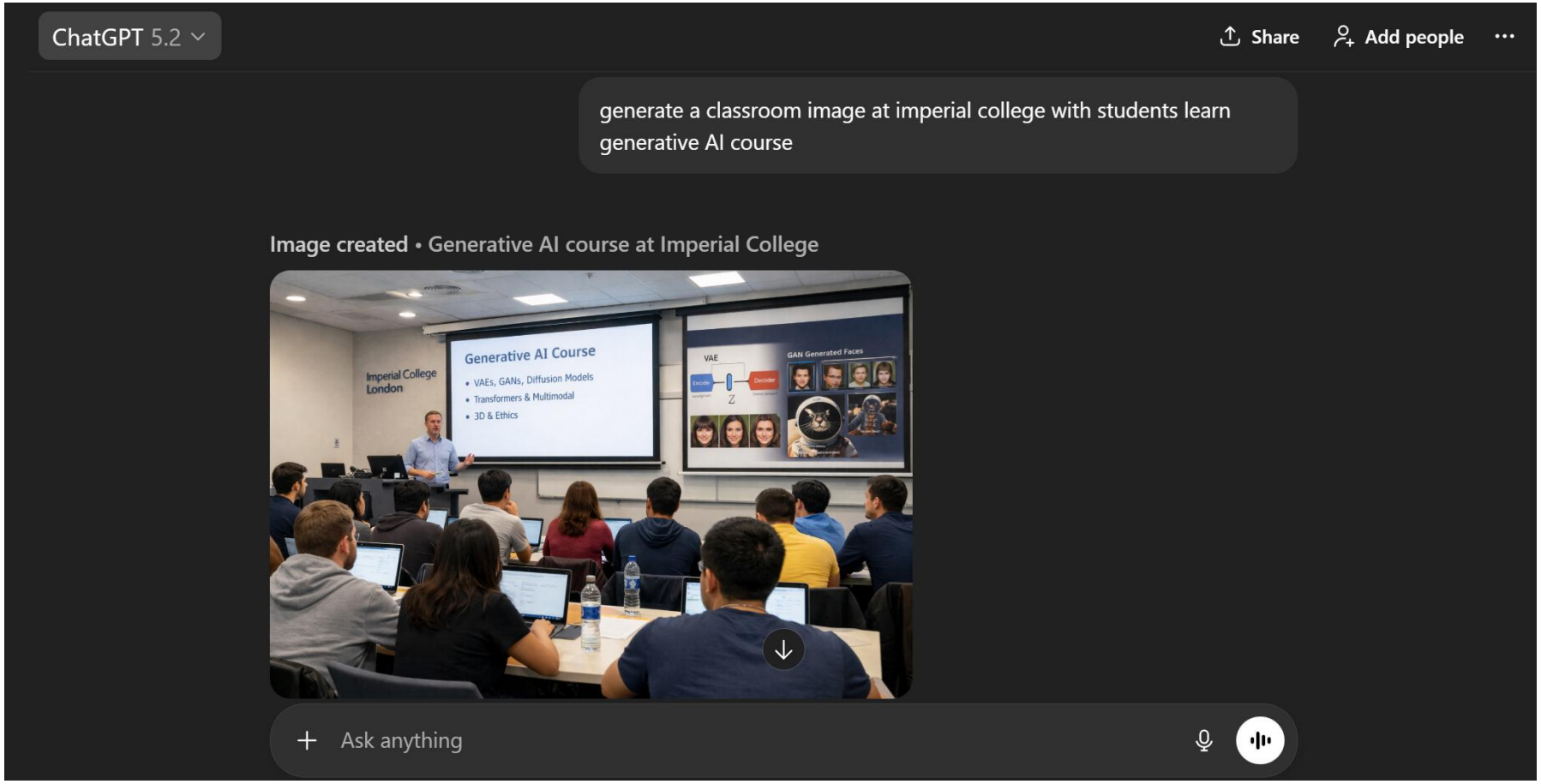
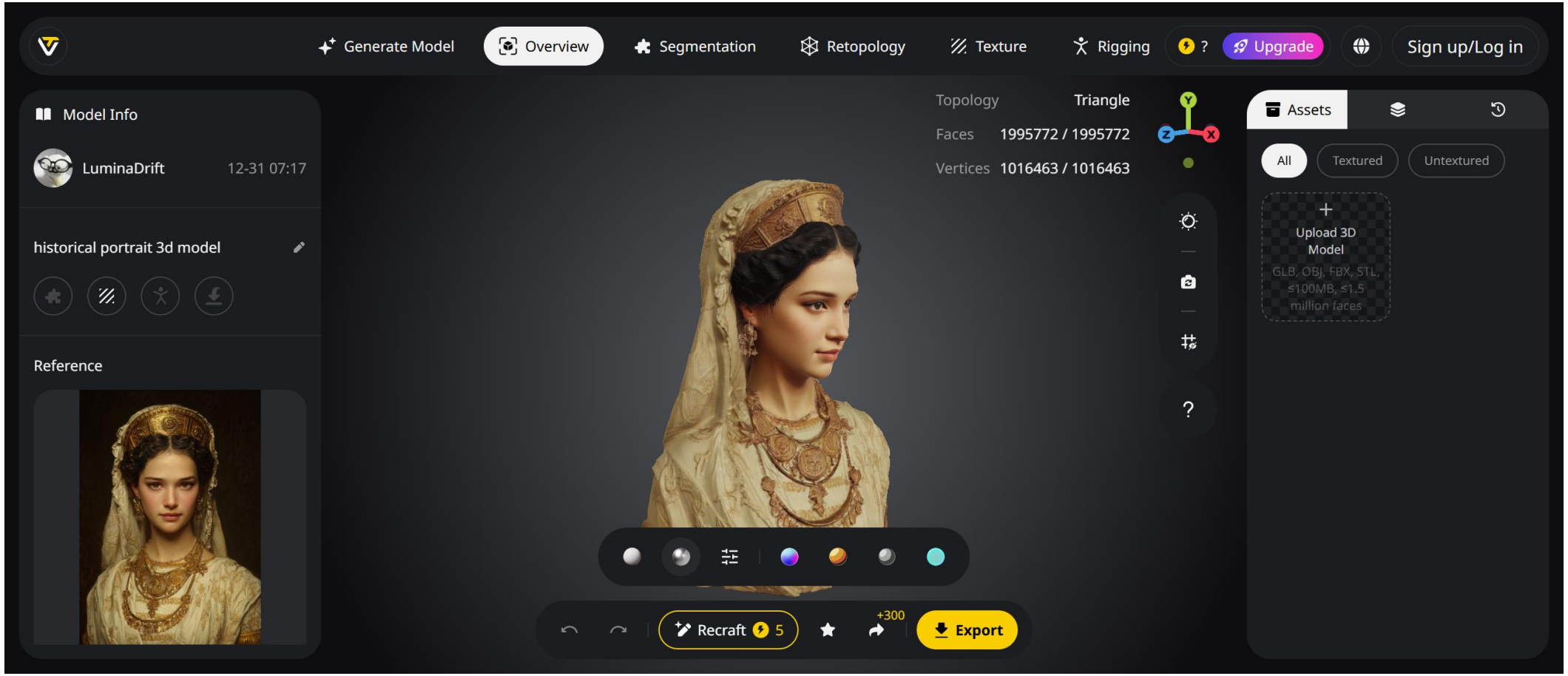


Image Generation

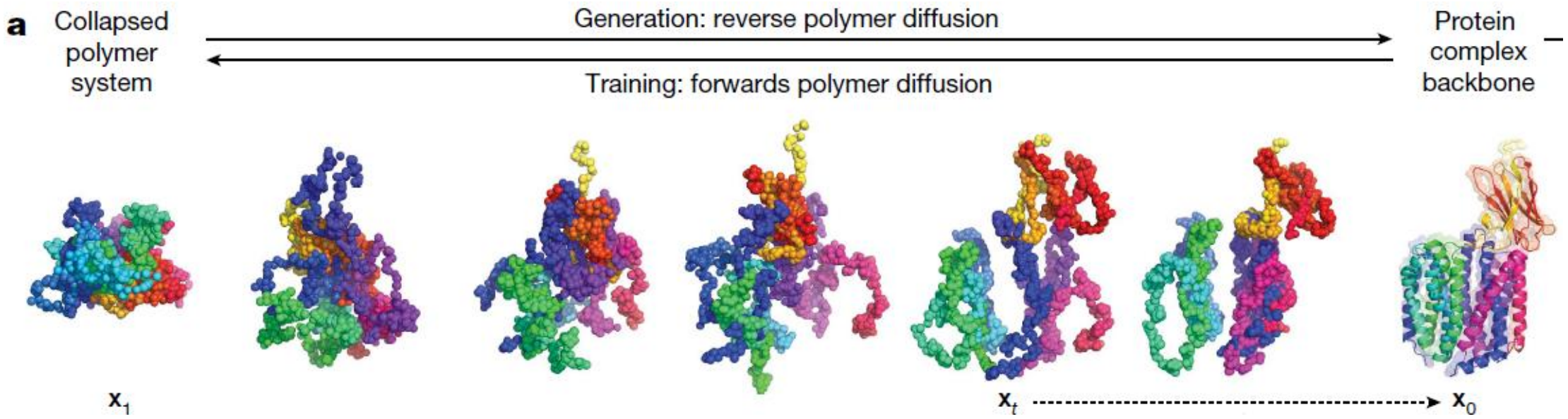


Video Generation

3D Generation

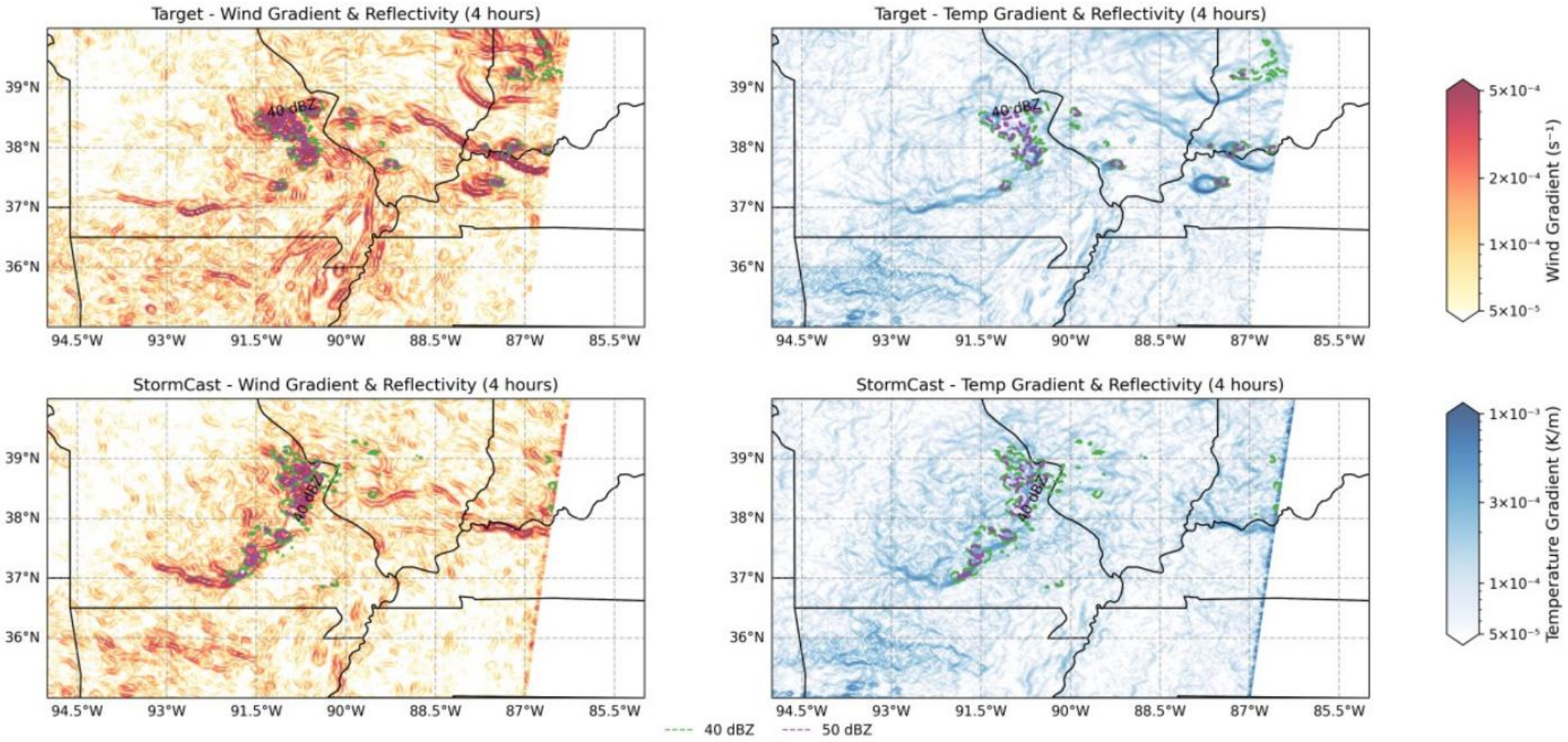


Protein Generation



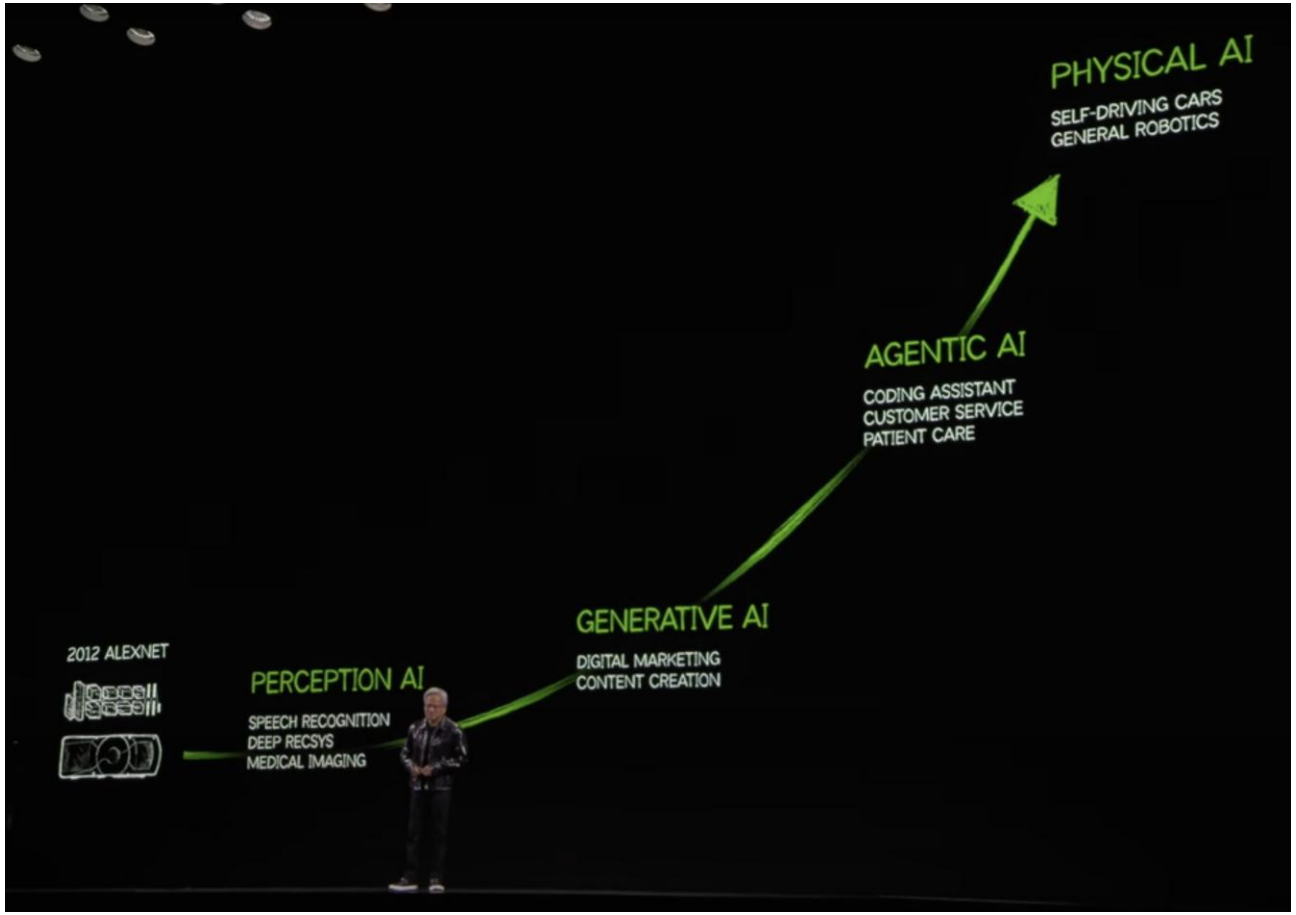
Highly accurate protein structure prediction with AlphaFold, Nature 2021
Illuminating protein space with a programmable generative model, Nature 2023

Weather Forecasting



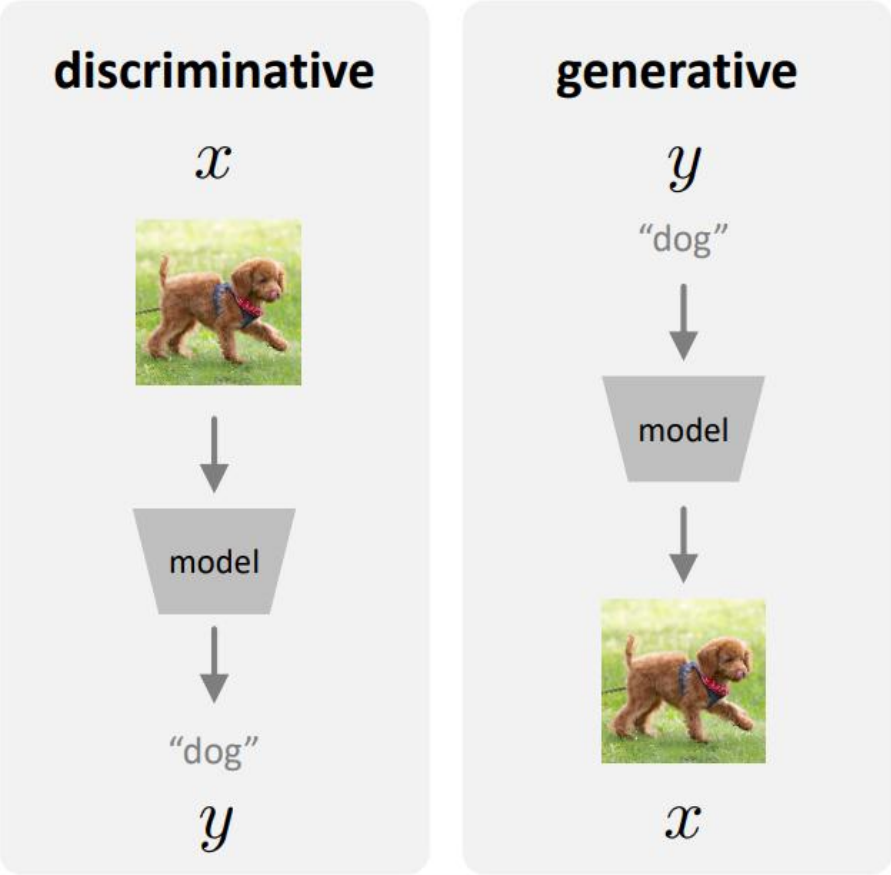
Kilometer-Scale Convection Allowing Model Emulation using Generative Diffusion Modeling, 2024

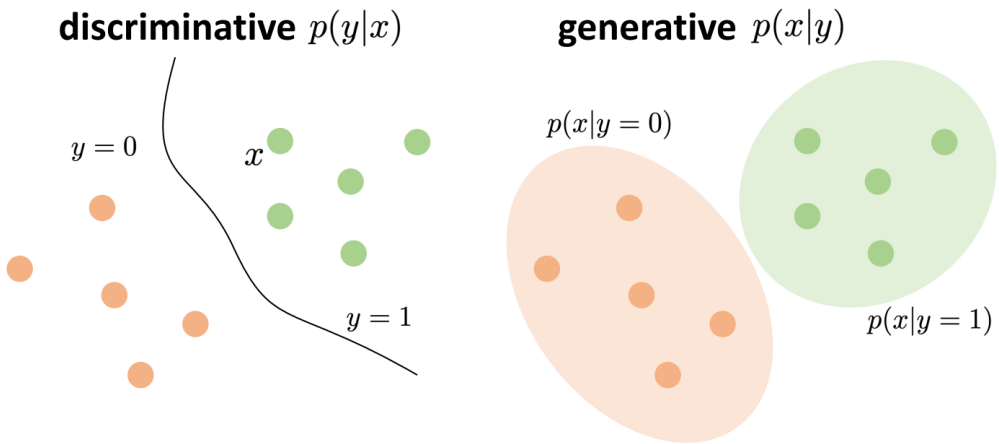
Development of AI



*Perception AI (Discriminative AI)->**Generative AI**->Agentic AI->Physical AI*

Discriminative vs. Generative Models





Discriminative

- *sample x -> label y*
- *One desired output*

Generative

- *label y -> sample x*
- *Many possible outputs*

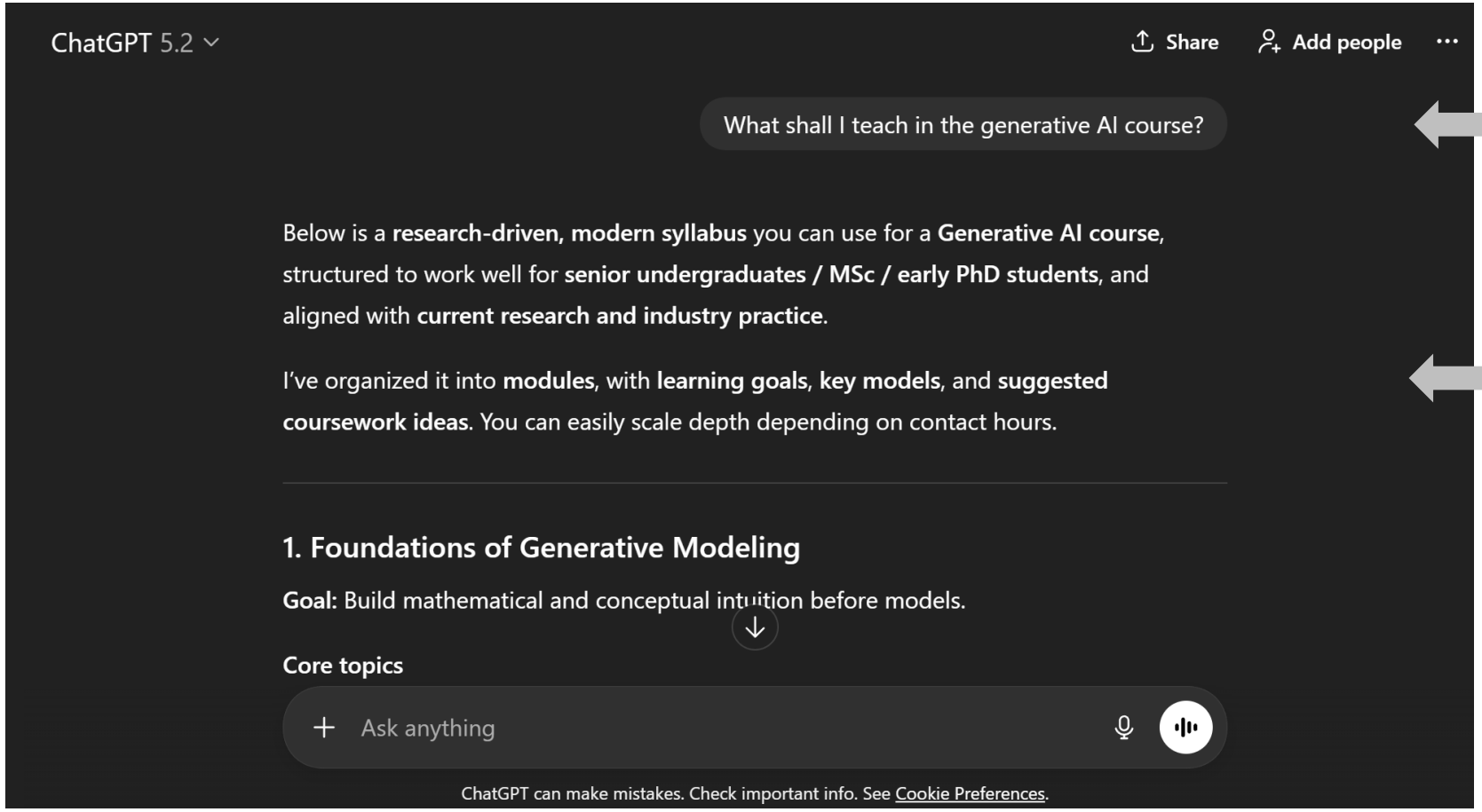
Formulating Problems as Generative Models

Core Objective: Modeling $p(x|y)$

Generative models aim to learn the conditional probability distribution of a high-dimensional output x given a specific input or context y.

Input Context (y)	**Generated Output (x)**
The "Prompt" or Context	**The "Result" or Data**
• **Condition:** Text prompts or settings	• **"Data":** High-dimensional content
• **Constraint:** Physical or logical bounds	• **Samples:** Specific instances
• **Labels:** Class categories (e.g., "Cat")	• **Observations:** Raw signal/pixels
• **Attributes:** Specific traits (e.g., "Blue")	• **Measurements:** Quantifiable data
Characteristics:	**Characteristics:**
• More **Abstract**	• More **Concrete**
• Less **Informative**	• More **Informative**

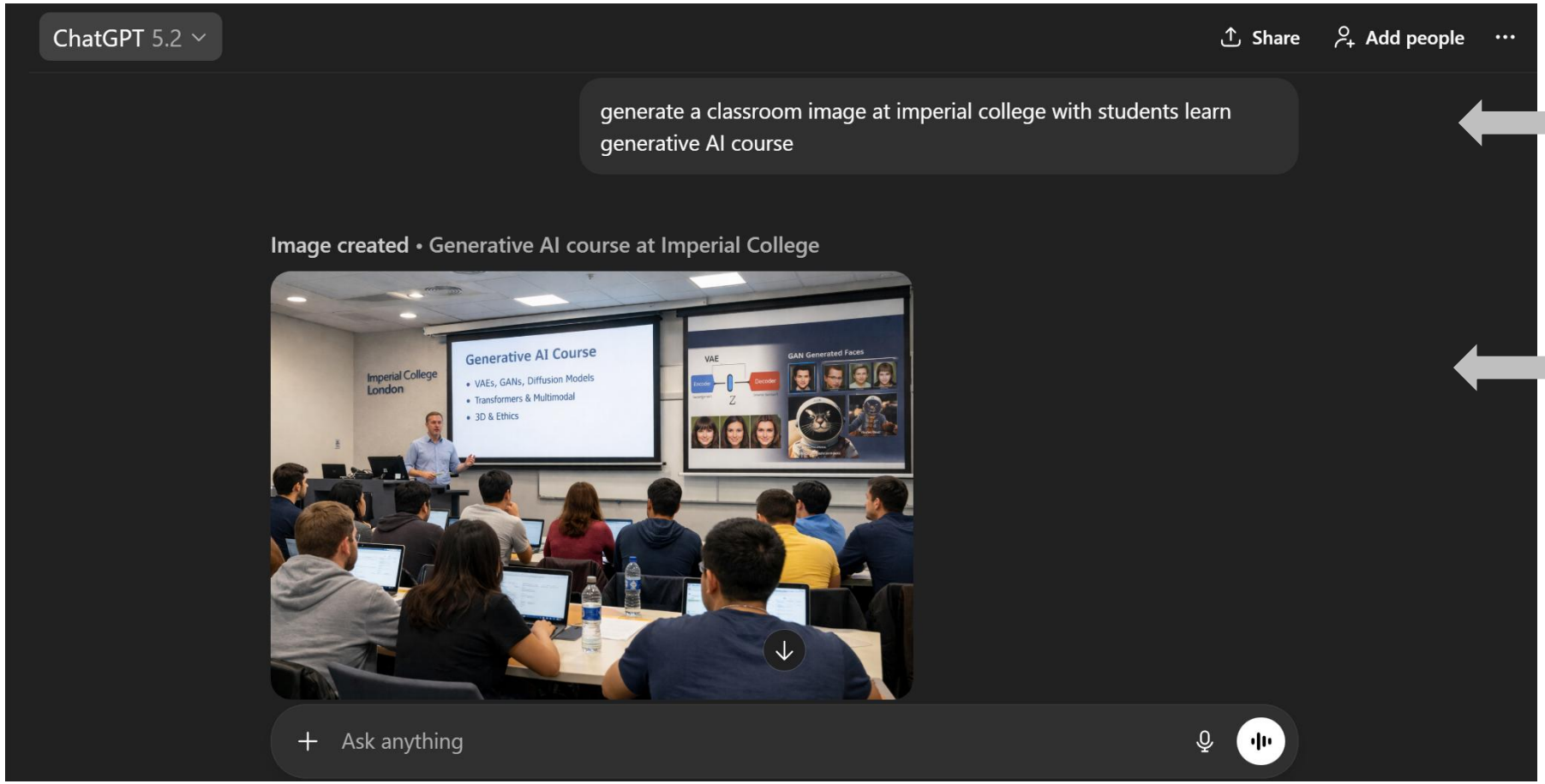
Case Study: Formulating as $p(x|y)$



y: text prompt

x: response of chatbot

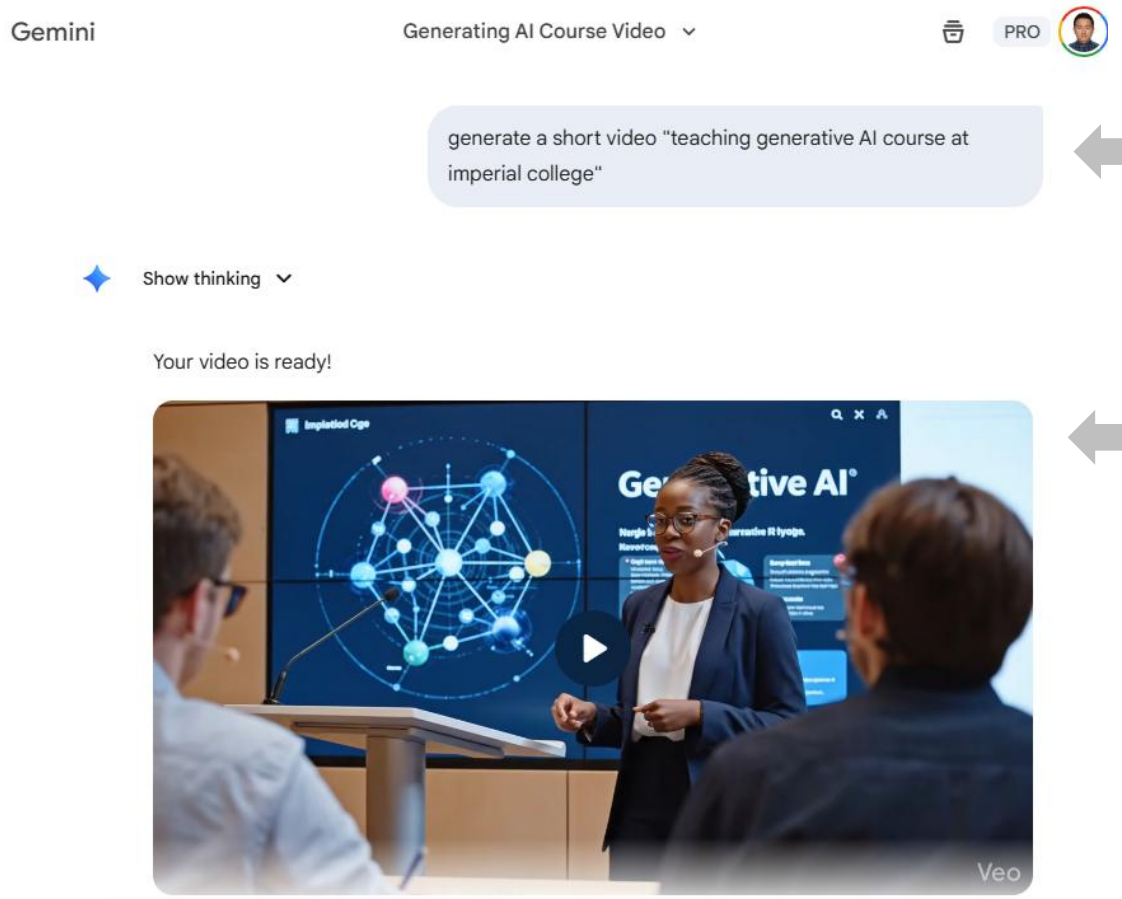
Case Study: Formulating as $p(x|y)$



y: text prompt

x: image

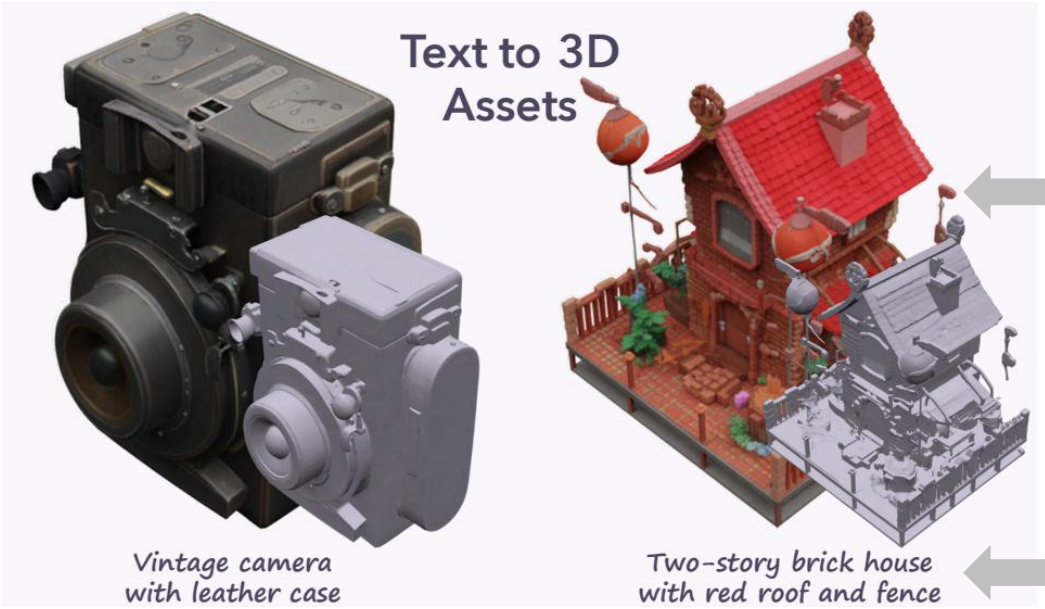
Case Study: Formulating as $p(x|y)$



y: text prompt

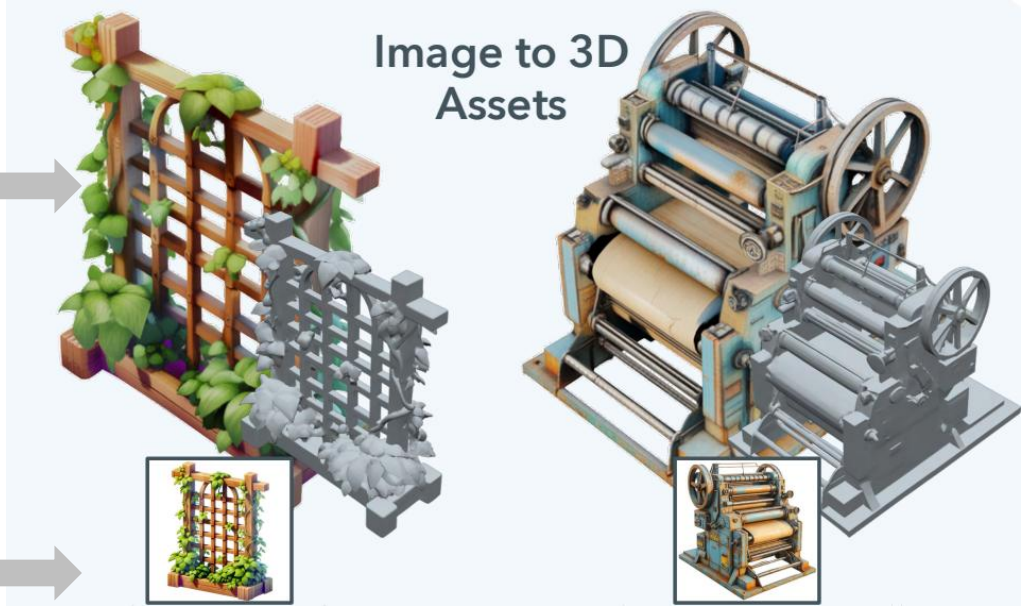
x: video

Case Study: Formulating as $p(x|y)$



x: 3D asset

y: prompt



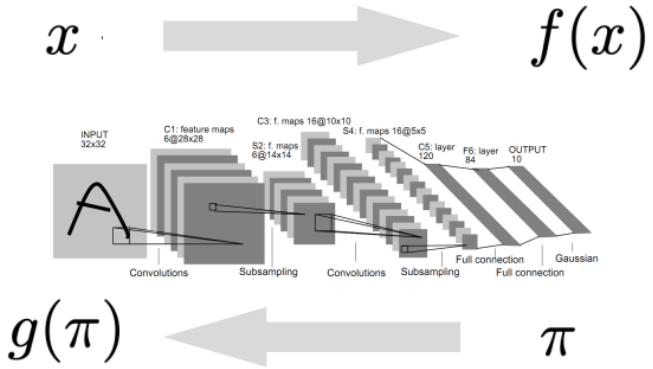
Structured 3D Latents for Scalable and Versatile 3D Generation, CVPR 2025

Learning to Represent Instances vs. Distributions

1. Learning to Represent Data Instances

This approach focuses on mapping individual data points to a descriptive feature space.

- **Transformation:** Map raw data x to a feature representation $f(x)$.
- **Optimization:** Minimize the loss function $\mathcal{L}(y, f(x))$ relative to a specific target y.
- **Visual Analogy:** Often represented by a feed-forward network (like a CNN) that extracts features from an input image.



2. Learning to Represent Probability Distributions

This approach focuses on generative modeling, transforming simple noise into complex, structured data.

- **Transformation:** Map a simple source distribution π (such as Gaussian or uniform noise) to a complex data distribution $g(\pi)$.
- **Optimization:** Minimize the loss function $\mathcal{L}(p_x, g(\pi))$ to align the generated distribution with the true data distribution p_x.
- **Directionality:** Acts as the inverse of feature extraction, moving from latent space back to the data domain.

Conceptual Overview

Feature	Instance Representation ($x \to f(x)$)	Distribution Representation ($\pi \to g(\pi)$)
Primary Goal	Feature Extraction / Classification	Data Generation / Synthesis
Input Type	Concrete Data (e.g., Image Pixels)	Latent Noise / Simple Distribution
Output Type	Abstract Features / Labels	Complex Data Samples
Loss Target	Specific Labels (y)	Data Distribution (p_x)

Probabilistic Modelling

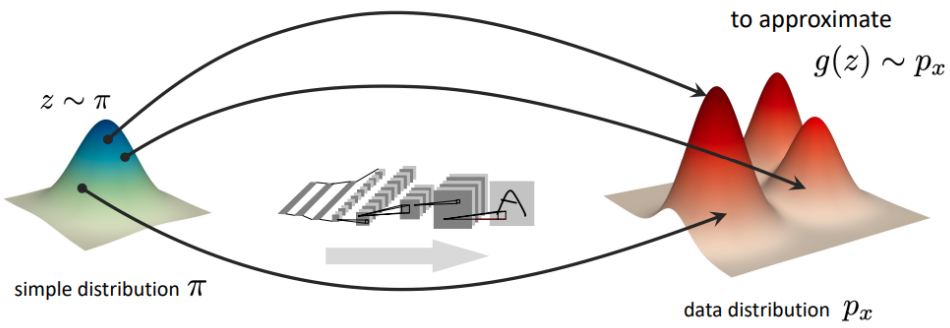
Where does probability come from?

- From **assumptions about the data generation process**
- Data is not arbitrary, it is produced by **latent causes** governed by probability distributions

We assume observations are generated through a **two-stage process**:

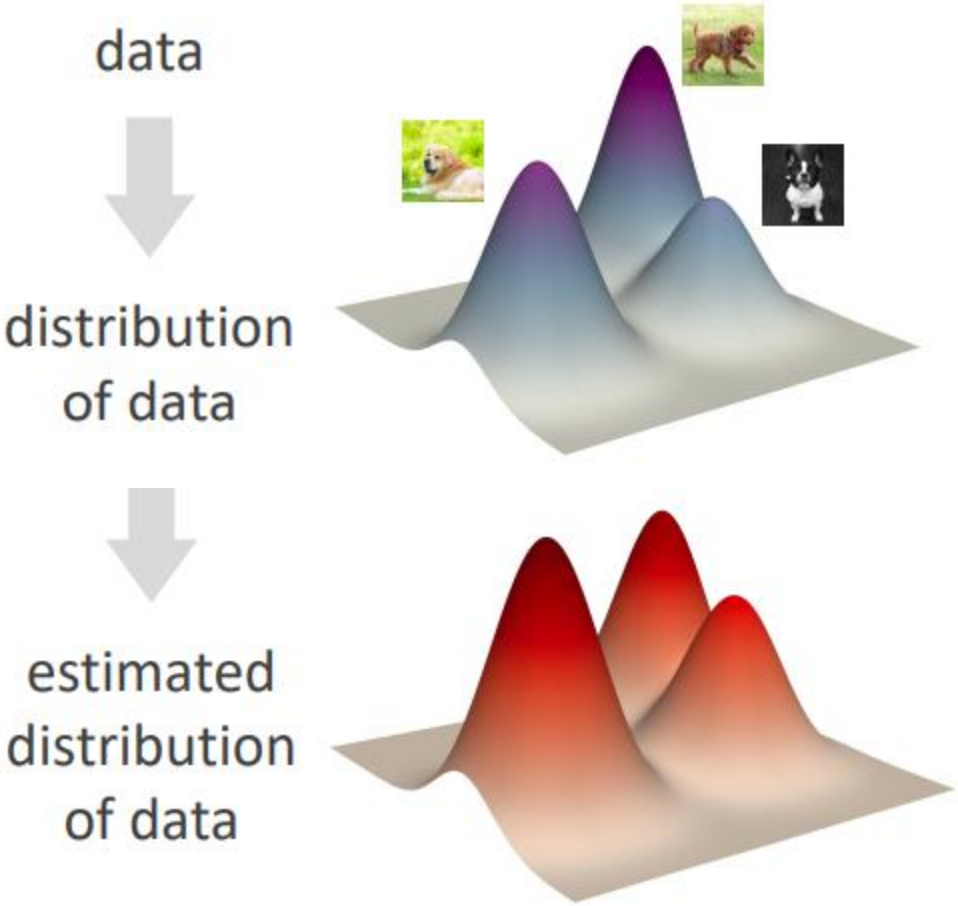
1. Latent variables are sampled from simple distributions
2. Observations are produced by transforming these latent variables through a complex function

- From simple to complex distributions

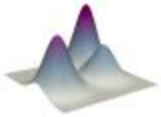
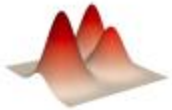


- Complexity in data arises from **simple latent variables + complex mappings**
- Learning = **inferring latent causes** or **approximating the generative process**

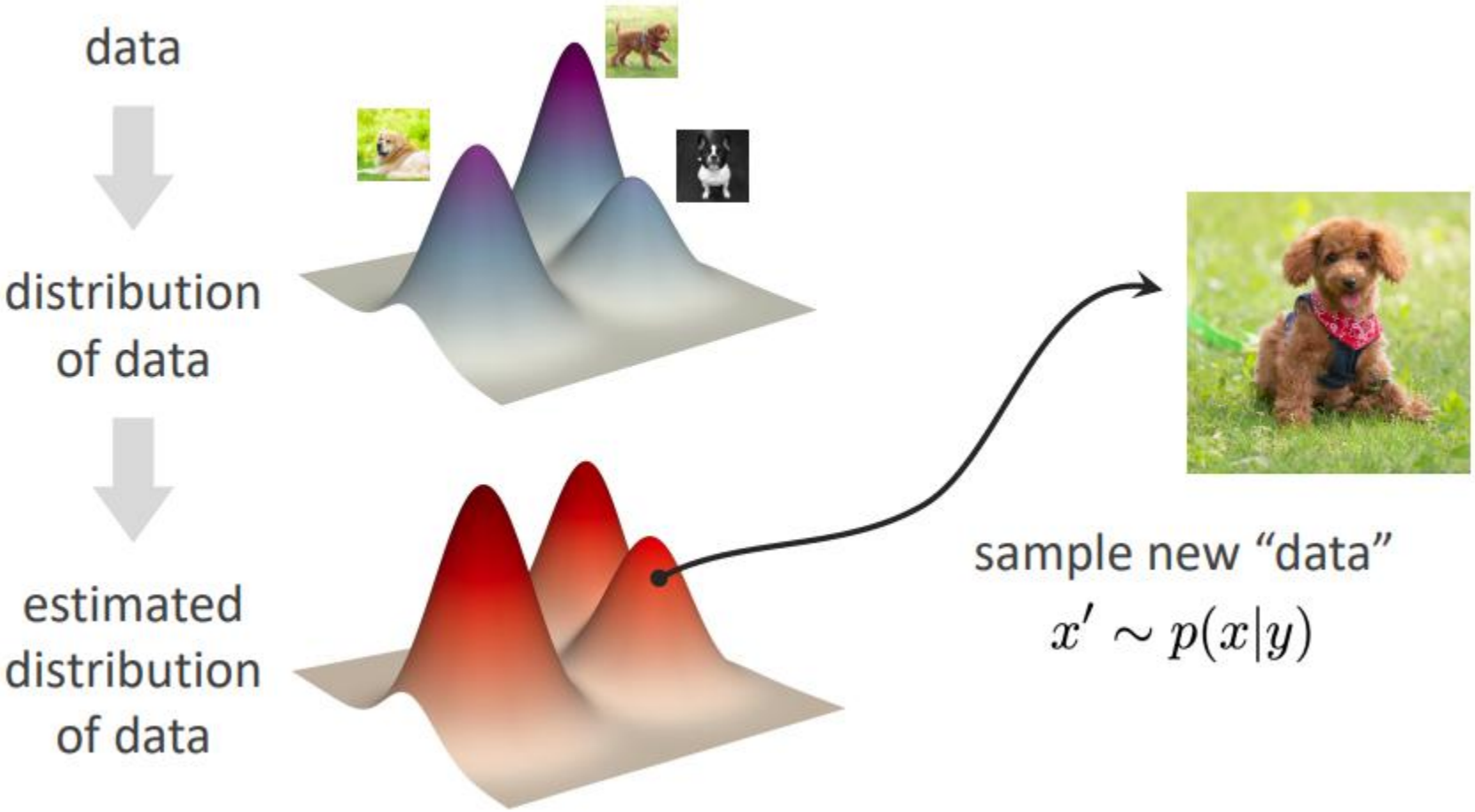
Probabilistic Modelling in Generative Models



- Optimize a loss function

$\mathcal{L}($  ,  $)$

Probabilistic Modelling in Generative Models

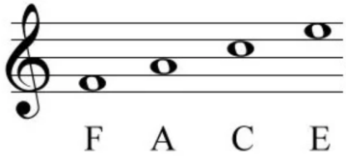


Types of Data Distributions

Generative models must account for the mathematical nature of the data they are modeling:

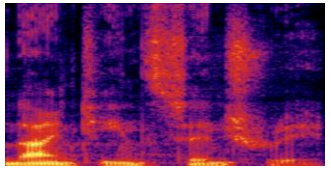
1. Discrete Distributions

- **Definition:** The data can only take on specific, countable values.
- **Mathematical Constraint:** The sum of all probabilities must equal 1: $\sum_i P(x_i) = 1$.
- **Examples:** Text (tokens), Music Notes.



2. Continuous Distributions

- **Definition:** The data can take on an infinite number of values within a range.
- **Mathematical Constraint:** The integral of the probability density function must equal 1: $\int_x p(x)dx = 1$.
- **Examples:** Images (pixel intensities), Audio waveforms.

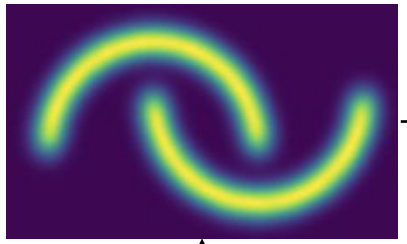


Generative Modeling

Data Distribution p

Training Dataset
$\{x_1, x_2, \dots, x_N\},\ x_i \sim p$

Model Distribution p_θ



Training Algorithm

Model θ

Sample

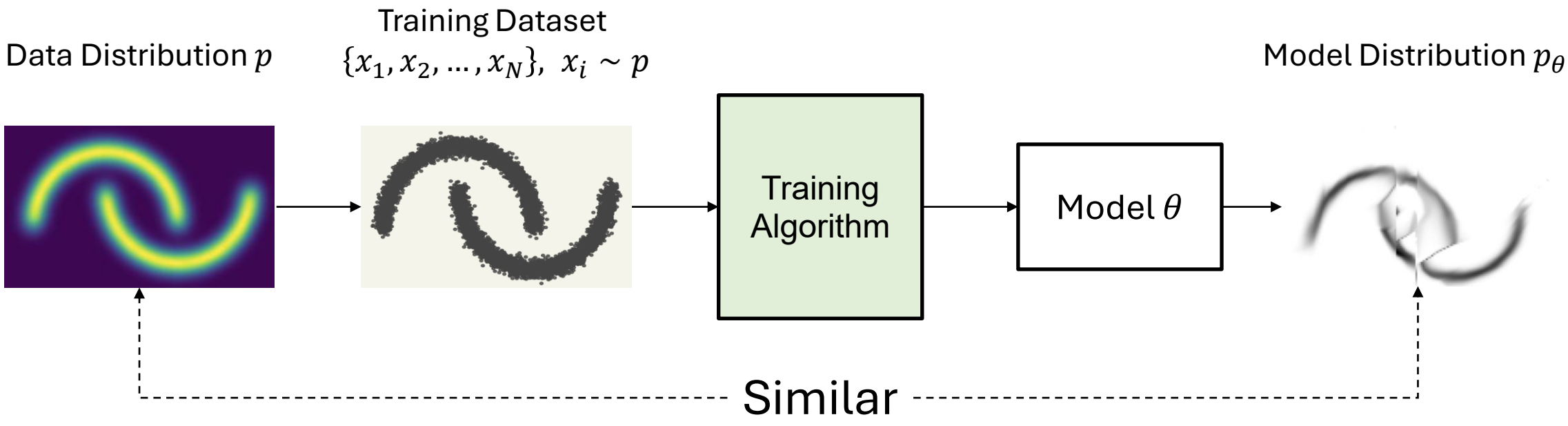
Similar

Estimating $p_\theta(x)$ and sample from $p_\theta(x)$ are often difference processes:

- Density estimation: Input x, output $p_\theta(x)$
- Generation/sampling: No input, output a sample $x' \sim p_\theta$

Figures from: https://dmol.pub/dl/flows.html

Generative Modeling



How to measure distance between distributions?

Kullback–Leibler (KL) Divergence

Kullback-Leibler (KL) Divergence is a fundamental tool used in generative modeling to measure how one probability distribution differs from a second, reference probability distribution.

The Mathematical Foundation

For discrete distributions, KL divergence is calculated as the sum of the probability of an event in the target distribution multiplied by the log of the ratio of the probabilities between the two distributions:

$$KL(p \parallel q) = \sum_i p(x_i) \log \frac{p(x_i)}{q(x_i)}$$

- $p(x_i)$**:** The "true" or target data distribution.
- $q(x_i)$**:** The approximated distribution (the model's output).

Key Properties to Note:

1. **Non-Symmetric:** $KL(p \parallel q) \neq KL(q \parallel p)$. The "distance" depends on which distribution is the reference.
2. **Always Non-Negative:** KL divergence is always ≥ 0. It is exactly 0 only if the two distributions are identical.

Example Calculation: KL Divergence

Example: Fair vs. Biased Coins

Consider two distributions representing coin flips:

- **Distribution p (Fair):** $(p_{\text{tail}}, p_{\text{head}}) = (0.5, 0.5)$
- **Distribution q (Biased):** $(q_{\text{tail}}, q_{\text{head}}) = (0.75, 0.25)$

- Q1: Compute $KL(p \parallel q)$ and $KL(q \parallel p)$
- Q2: For each KL divergence, is it positive or negative?

Example Calculation: KL Divergence

Q1: Computing the Scores

Using the formula $KL(p \parallel q) = \sum_i p(x_i) \log \frac{p(x_i)}{q(x_i)}$:

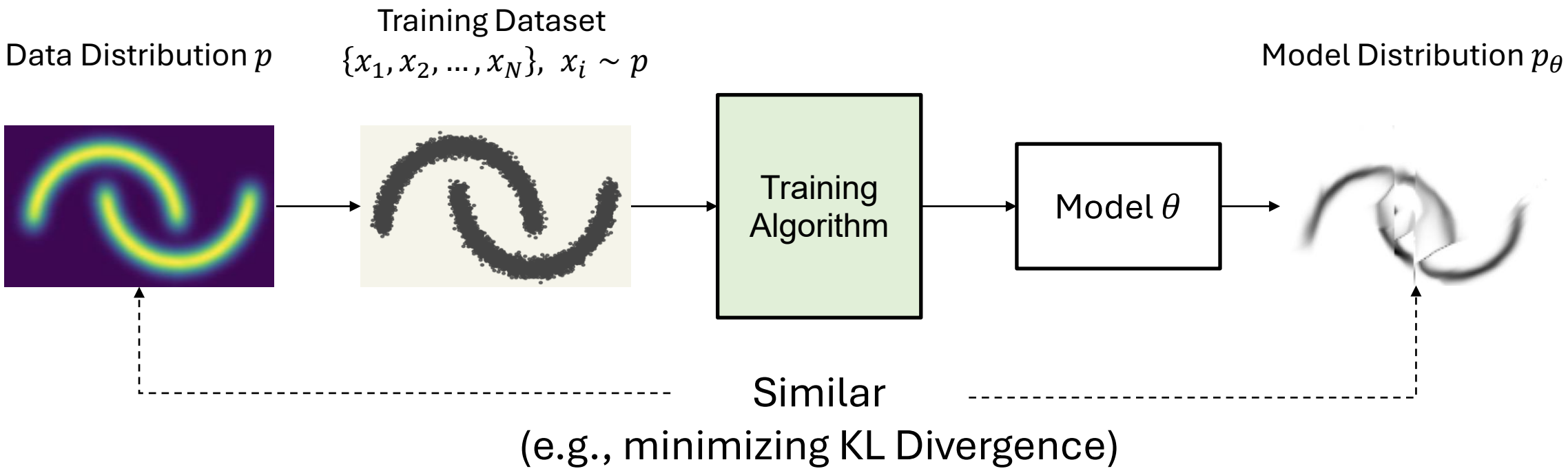
1. $KL(p \parallel q)$ **(Fair relative to Biased):**
   - $= 0.5 \ln\left(\frac{0.5}{0.75}\right) + 0.5 \ln\left(\frac{0.5}{0.25}\right)$
   - $= 0.5 \ln\left(\frac{2}{3}\right) + 0.5 \ln(2)$
   - $\approx \mathbf{0.1438}$
2. $KL(q \parallel p)$ **(Biased relative to Fair):**
   - $= 0.75 \ln\left(\frac{0.75}{0.5}\right) + 0.25 \ln\left(\frac{0.25}{0.5}\right)$
   - $= 0.75 \ln(1.5) + 0.25 \ln(0.5)$
   - $\approx \mathbf{0.1308}$

Example Calculation: KL Divergence

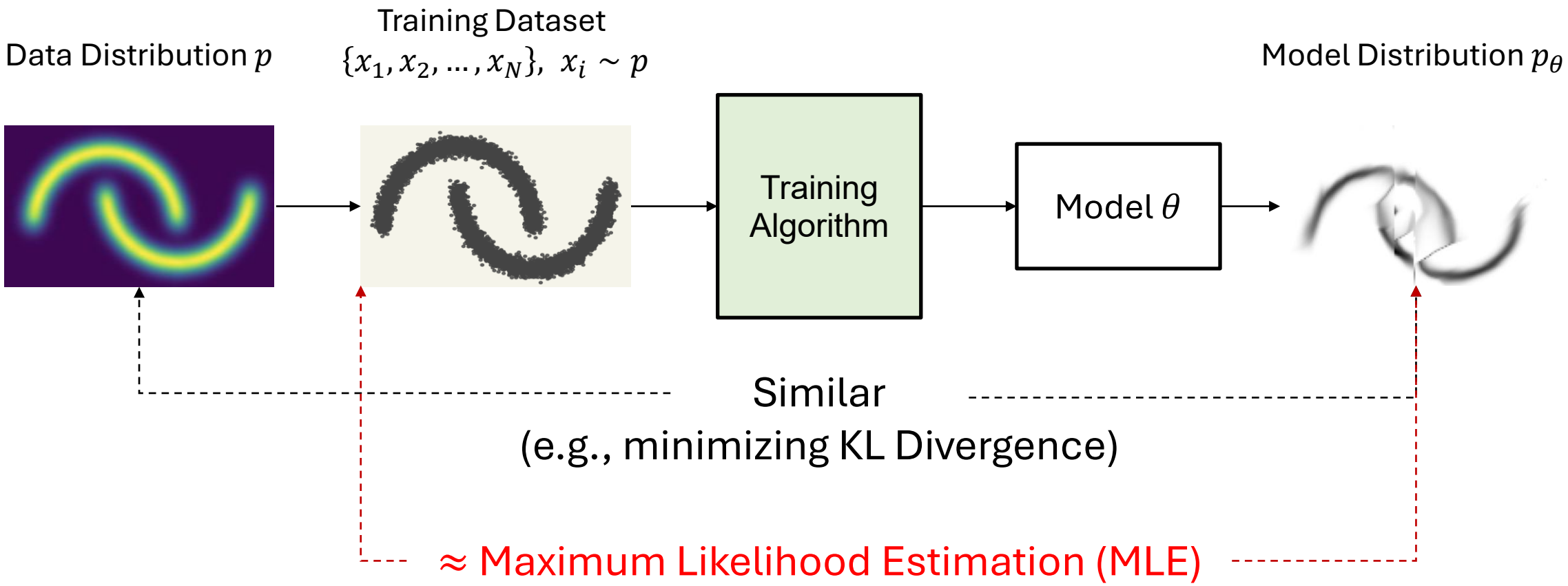
Q2: Analysis of Results

- **Is it positive or negative?**
  - Both scores are **positive**.
  - **Reason:** According to Gibbs' inequality, KL divergence is always ≥ 0. It is only 0 if $p = q$. Since these distributions differ, the scores must be positive.
- **Observation on Symmetry:**
  - Notice that $0.1438 \neq 0.1308$. This demonstrates that KL divergence is **non-symmetric**, meaning the "information loss" depends on which distribution is treated as the true reference.

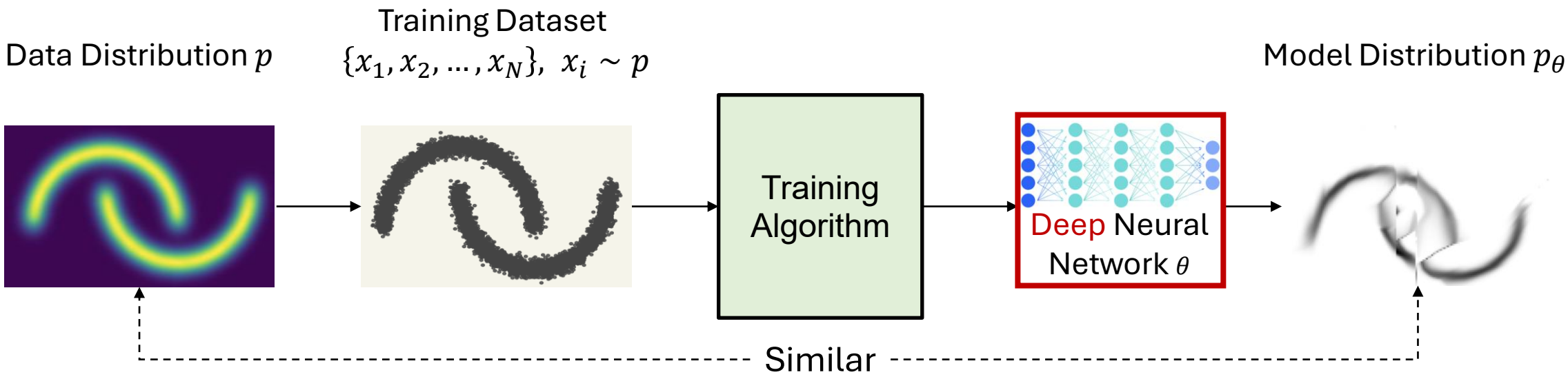
Generative Modeling

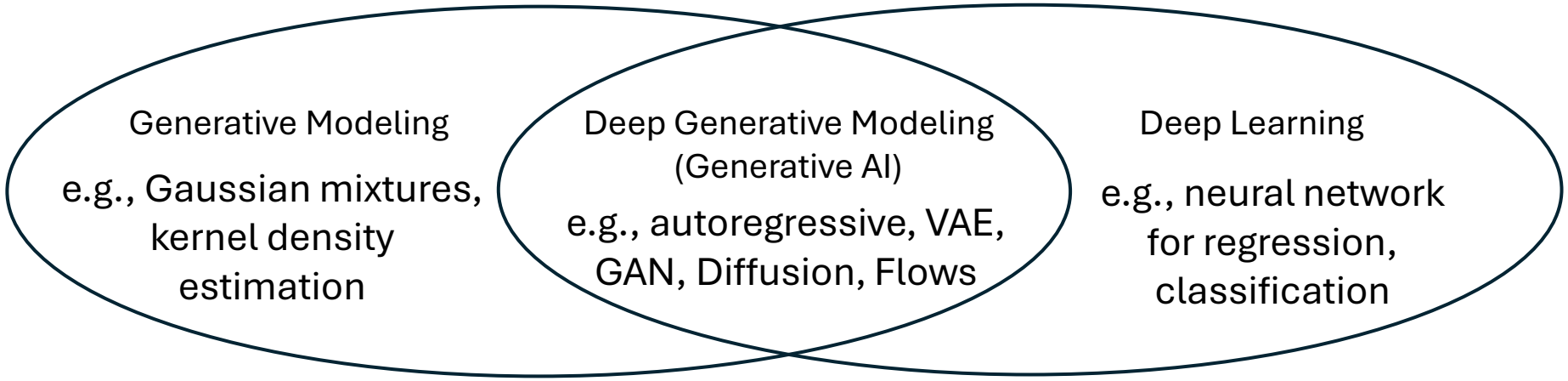


Generative Modeling

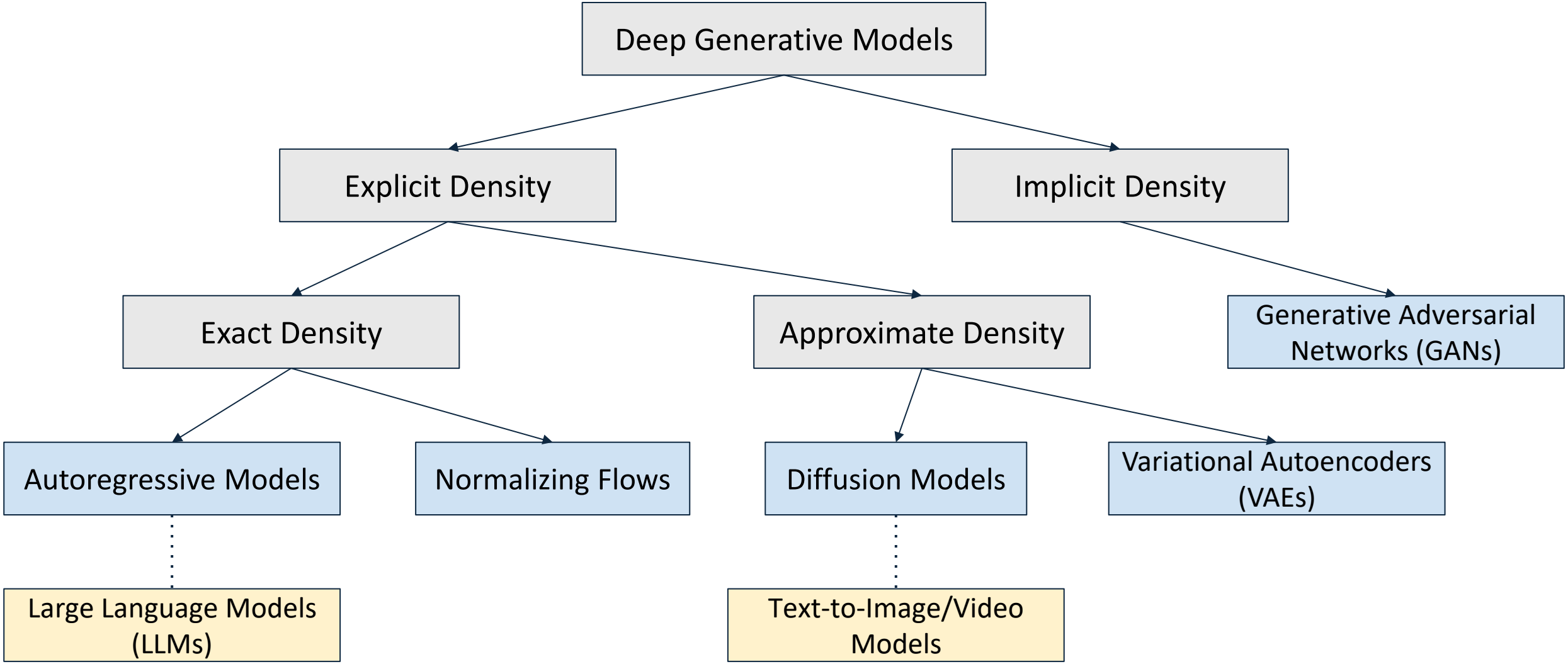


Deep Generative Modeling





Deep Generative Models



Prerequisites

- Basic knowledge about deep learning and machine learning.
- Proficiency in some programming language, preferably Python (Pytorch), required.
- Basic knowledge of probabilities and calculus:
  - Gradients, gradient-descent optimization, backpropagation
  - Random variables, independence, conditional independence
  - Bayes rule, chain rule, change of variables formulas

Syllabus

Date	Weekday	No.	Topic	Lecture Notes
1/13/2026	Tuesday	1	**Introduction to Generative AI & Preliminary: Basics of Deep Learning**	Lecture1 : Jiankang Deng
1/16/2025	Friday		Tutorial: Basics of Deep Learning	Tutorial & Lab: Yanzuo Lu
1/20/2026	Tuesday	2	**Text Generation**	Lecture 2: Jiankang Deng
1/23/2026	Friday		Tutorial: Text Generation	Tutorial & Lab: Yanzuo Lu
1/27/2026	Tuesday	3	**Image Generation 1**	Lecture 3: Jiankang Deng
1/30/2026	Friday		Tutorial: Image Generation 1	Tutorial & Lab: Yanzuo Lu
2/3/2026	Tuesday	4	**Image Generation 2**	Lecture 4: Jiankang Deng
2/6/2026	Friday		Tutorial: Image Generation 2	Tutorial & Lab: Yanzuo Lu
2/10/2026	Tuesday	5	**Audio Generation**	Lecture 5: Ronglai Zuo
2/13/2026	Friday		Tutorial: Audio Generation	Tutorial & Lab: Ronglai Zuo
2/17/2026	Tuesday	6	**3D Generation**	Lecture 6: Rolandos Alexandros Potamias
2/20/2026	Friday		Tutorial: 3D Generation	Tutorial & Lab: Zerui Chen
2/24/2026	Tuesday	7	**Video Generation**	Lecture 7: Jiankang Deng
2/27/2026	Friday		Tutorial: Video Generation	Tutorial & Lab: Yanzuo Lu
3/1/2026			Project Deadline	
3/3/2026	Tuesday	Guest Lecture: Runway		
3/6/2026	Friday	Guest Lecture: Google/Meta		

Logistics – Grading Policies

- **Course Project (30%)**
  - **Teams:** Maximum of 3 students per team. All members receive a collective score. Group registration Link.
  - **Hugging Face Demo (20%):** Development and deployment of a functional demo.
  - **Poster (10%):** Visual presentation of project motivation, methodology, and results.

- **Final Exam (70%)**
  - **Format:** Focuses on core concepts and practical calculations.
  - **Resources:** Please note that past exam papers are not provided.
  - **Review Session:** A dedicated revision period will take place during the final week (March 3).

Course Projects

- **1. Innovation**
- **The "Creative" Challenge:** Choose a topic that excites you. Your project should be polished and unique enough to theoretically attract real-world users or subscribers.

- **2. Core Technical Requirements** Your workflow must involve at least one of the following generative AI components:
- Develop and train a small model from scratch.
- Take a pre-trained model and adapt it for a niche, creative application.

- **3. Some Project Ideas**
- **Image Restoration & Enhancement:** Go beyond basic inpainting or super-resolution to build a tool that restores vintage or damaged photographs.
- **Multimodal Synthesis:** Develop a system that generates synchronized audio or music from text prompts and visual cues.
- **Creative Text Generation:** Fine-tune a model to emulate the prose of historical figures or to generate functional code for specialized programming languages.

Late Policy

Submission Deadline

- **Deadline: Sunday, March 1, 2026, at 11:59 PM GMT**.
- **Strict Cut-off:** Late submissions will not be accepted.
- **Requirements:** Upload a PDF version of your poster. Ensure the PDF includes a functional link to your project demo.

Extenuating Circumstances

- If you are facing serious medical issues or major life events, please submit an extension request through the Department. We will coordinate with them to determine a suitable resolution.

GenAI Tool Policy

Labs & Tutorials: You may NOT use GenAI to generate full solutions or copy-paste answers. However, you are encouraged to use these tools for creative inspiration and to help troubleshoot or explain errors.

Project Development: GenAI tools are permitted for use during project development. Utilize them to ensure your system is functional and to enhance the visual appeal and professional quality of your poster.

Final Exam

Exam format
Closed-book, written exam
Duration: 90 minutes

Structure
- Analytical derivations
- Conceptual understanding
- Short design / reasoning questions
- Multiple-choice (with justification where required)

Format and difficulty are similar to COMP70010 (Deep Learning)

Typical Question Styles
- ✏️ Derive an objective
- 🧠 Explain why a design choice works or fails
- 🎛️ Design a generative model given constraints
- 📐 Compute dimensions, parameters, or complexity
- ✅ Select & justify correct statements

Indicative Topic Coverage
Students should be comfortable with both theory and practical reasoning across:
- Probabilistic generative models (VAEs, ELBO derivation, latent variable modelling)
- Implicit generative models (GANs, adversarial objectives, conditional generation)
- Diffusion & score-based models (Forward/reverse processes, noise schedules, sampling)
- Autoregressive & sequence models (Transformers, attention, likelihood factorisation)

Logistics

- Reading materials will be provided at the end of each lecture.
- Lab & Tutorial Session (Students are required to bring their own laptops. Exercises will be conducted using Google Colab.)
  - **Oliver Lu (ol125)**
  - **Ronglai Zuo**
  - **Zerui Chen**
- Teaching Assistants (Lecture Q&A and Course Project Q&A):
  - **Sherwin Vadakkumcherry (svadakku)**
  - **Yura Choi (yc825)**
  - **Ruomeng Fan (rf221)**
  - **Lapo Rastrelli (lr4617)**
  - **Jacob Si (js2723)**
  - **Junhao Song (js3623)**
  - **Xavier Sumba Toral (xxs22)**

Recap

- Application & formulation of generative models
- KL divergence
  - Properties (non-negative and asymmetric)
  - Minimizing KL = Maximizing likelihood
- *Deep* generative models
  - Challenges
  - Taxonomy